Johnson Yep

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Education

University of Waterloo — Waterloo, Ontario
Mechatronics and Robotics Engineering — April 2030
Relevant Courses: OOP in C++, Data Structures and Algorithms, Linear Algebra, Circuits

Skills

- **Languages:** Python, C++, Java
- **Perception & ML:** OpenCV, PyTorch, Ultralytics YOLO, Roboflow, MediaPipe, Point Clouds, Hand-Eye Camera Calibration, Homography
- **Robotics:** ROS2, MoveIt, Isaac Lab, Isaac Sim, Foxglove, Universal Robots UR20
- **Developer Tools:** Git, Docker, Linux, GitHub Actions
- **Controls & CAD:** Allen-Bradley PLCs, Ladder Logic, EtherNet/IP, HMI, SolidWorks, AutoCAD, GD&T

Experience

Robotics & Software Engineering Intern — Sept 2026 - Present
RoBIM Technologies — View Work

- Developed vision software for modular robotic cells that prefabricate structural components for construction.
- Built an RGB-D lumber survey pipeline with **ROS 2** and **MoveIt** that finds boards anywhere in the workspace and scans them into **3D point clouds**; cut scan time 2.5×.
- Developed a **MoveIt** pick-and-place system that sorts boards by length using collision-free **motion planning**, and designed its gripper attachment in **SolidWorks**.
- Fine-tuned a **YOLO** instance-segmentation model on a custom dataset labelled in **Roboflow**, reaching 0.836 mask mAP50-95 on held-out images.

Robotics Engineering Intern — Jan 2026 - May 2026
Toyota Boshoku — View Work

- Built a real-time rack-occupancy vision system in **Python** and **OpenCV** with lens distortion correction, homography top-down warping, per-cell baselines, and multithreaded RTSP capture, signalling **Allen-Bradley PLCs** over EtherNet/IP with a heartbeat failsafe.
- Integrated the system with **MiR AMR** missions to automate transport of **100**+ racks per shift.
- Deployed **Keyence** vision cameras for defect detection, preprocessing and labeling image datasets to train classification models that reduced defective parts reaching downstream processes.

Humanoid Robotics Team — May 2026 - Present
WATonomous — View Work

- Trained a **PPO** policy in **Isaac Lab** for an SO-ARM101 arm to push objects through contact-only manipulation, using **11** shaped reward terms and a performance-gated curriculum to reach **85%** success on fully randomized spawns.
- Built a **32**×**16**-taxel open-source tactile sensor with firmware, a real-time visualization pipeline, and a simulation environment.

Driveline Team — Sep 2025 - May 2026
uWaterloo Formula Electric — View Work

- Designed and manufactured drilling and motor-mounting jigs in **SolidWorks** with **GD&T** drawings, machining features to ±**0.1 mm**.

Projects

Autonomous Checkers Robot *C++* — View Project

- Built an autonomous robot that detects board pieces, selects moves, and physically moves the pieces.

Hand Tracking Computer Vision Guitar *Python, OpenCV, MediaPipe* — View Project

- Converted real-time hand and finger tracking into playable chords and notes.

Space Race *Java, CSS* — View Project

- Developed a 2D shooter game from scratch, designing gameplay, UI, and architecture end-to-end.